LOGISTIC REGRESSION

In [3]:
```
import pandas as pd
data=pd.read_csv("D:\Machine_Learning\Iris.csv")
data
```

Out[3]:

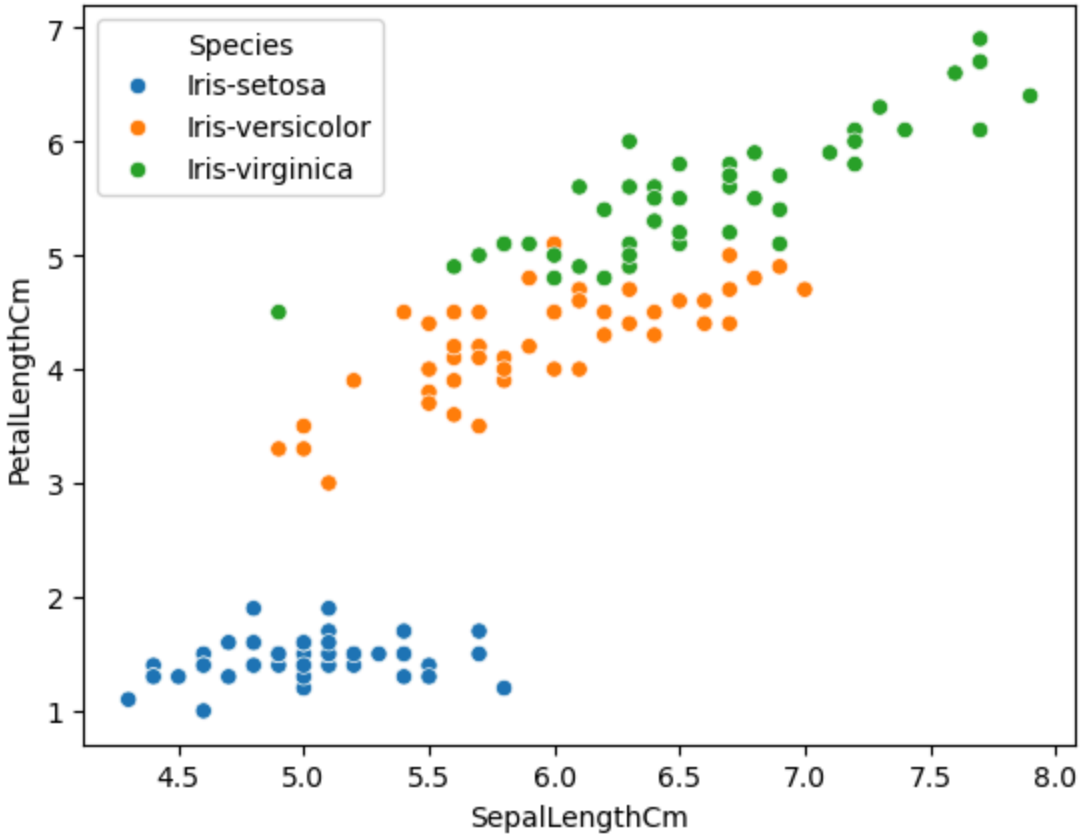
	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 6 columns

In [4]:
```
# DATA VISUALIZATION
import seaborn as sn
```

In [5]:
```
sn.scatterplot(x="SepalLengthCm",y="PetalLengthCm",data=data,hue="Species")
```

Out[5]:
```
<Axes: xlabel='SepalLengthCm', ylabel='PetalLengthCm'>
```



In [6]:
```
# LOGISTIC REGRESSION
# STEP 1: ASSIGNING VARIABLES(INDEPENDENT/DEPENDENT)

# INDEPENDENT VARIABLE
x=data[["SepalLengthCm","SepalWidthCm","PetalLengthCm","PetalWidthCm"]]

# DEPENDENT VARIABLE
y=data[["Species"]]
```

In [7]:
```
# STEP 2:IMPORT TRAIN TEST SPLIT
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.3)
```

In [8]:
```
print("total data: ",data.shape)
print("x_train data: ",x_train.shape)
print("x_test data: ",x_test.shape)
print("y_train data: ",y_train.shape)
print("y_test data: ",y_test.shape)
```

```
total data:  (150, 6)
x_train data:  (105, 4)
x_test data:  (45, 4)
y_train data:  (105, 1)
y_test data:  (45, 1)
```

In [9]:
```
# STEP 3: APPLY LOGISTIC REGRESSION
from sklearn.linear_model import LogisticRegression
model=LogisticRegression()
model
```

Out[9]:
```
LogisticRegression
▶ Parameters
```

In [10]:
```
# STEP 4 :FIT TO DATA
# Use only train(x_train,y_train)
model.fit(x_train,y_train)
```

```
C:\Users\Admin\anaconda3\Lib\site-packages\sklearn\utils\validation.py:1406: DataCon
versionWarning: A column-vector y was passed when a 1d array was expected. Please ch
ange the shape of y to (n_samples, ), for example using ravel().
  y = column_or_1d(y, warn=True)
C:\Users\Admin\anaconda3\Lib\site-packages\sklearn\linear_model\_logistic.py:473: Co
nvergenceWarning: lbfgs failed to converge after 100 iteration(s) (status=1):
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT

Increase the number of iterations to improve the convergence (max_iter=100).
You might also want to scale the data as shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
  n_iter_i = _check_optimize_result(
```

Out[10]:
```
LogisticRegression
▶ Parameters
```

In [11]:
```
# CHECK ACCURACY
model.score(x_test,y_test)
```

Out[11]:
```
0.9333333333333333
```

In [12]:
```
# PREDICTION OF NEW DATA
# IMPORT NUMPY FIRST(DATA MUST IN ARRAY FORM)
import numpy as np
data=np.array([[6.4,3.0,4.5,2.1]])
data
```

Out[12]:
```
array([[6.4, 3. , 4.5, 2.1]])
```

In [13]:
```
prediction=model.predict(data)
prediction
```

```
C:\Users\Admin\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWar
ning: X does not have valid feature names, but LogisticRegression was fitted with fe
ature names
  warnings.warn(
```

Out[13]:
```
array(['Iris-virginica'], dtype=object)
```